

Motion Detection Sensor using Arduino and PIR Sensor

A motion detection sensor is an electronic device used to detect movement in a specific area. In this project, a PIR (Passive Infrared) sensor is connected with an Arduino UNO to detect human motion. When movement is detected, LEDs glow and a buzzer produces sound.

How Does a Motion Detection Sensor Work?

The PIR sensor detects infrared radiation emitted by human bodies or moving objects. When a person moves in front of the sensor, the infrared energy changes. The PIR sensor sends a signal to the Arduino UNO. The Arduino processes this signal and activates connected components like LEDs and buzzers.

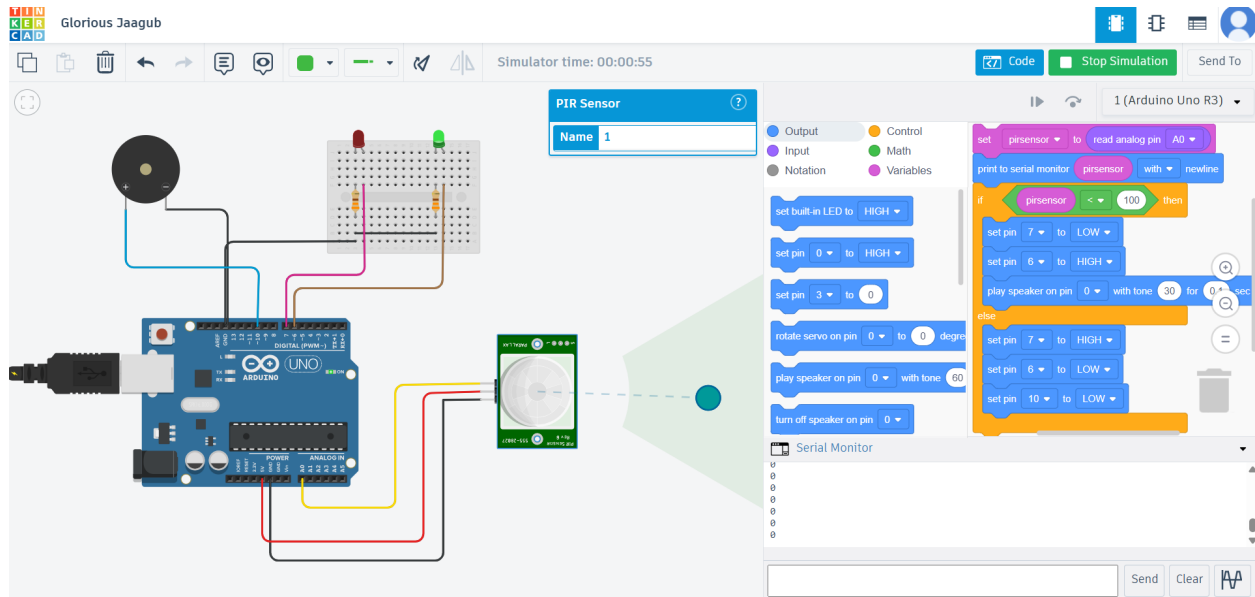
Main Components Used

Component	Purpose
Arduino UNO	Controls the entire system
PIR Sensor	Detects motion
LEDs	Indicate motion detection
Buzzer	Produces alarm sound
Breadboard & Wires	Circuit connections

Working Principle

1. The PIR sensor continuously monitors the surrounding area.
2. If no movement is detected, the green LED remains ON.
3. When motion is detected, the PIR sensor sends HIGH signal to Arduino.
4. The Arduino turns ON the red LED and buzzer alarm.
5. The system displays sensor values through Serial Monitor.

Circuit Simulation in Tinkercad



Arduino Code for Motion Detection Sensor

```
int pirsensor = 0;

void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(10, OUTPUT);
}

void loop()
{
  pirsensor = analogRead(A0);
  Serial.println(pirsensor);
  if (pirsensor >= 100) {
    digitalWrite(7, LOW);
    digitalWrite(6, HIGH);
    tone(10, 92, 100); // play tone 30 (F#2 = 92 Hz)
  } else {
    digitalWrite(7, HIGH);
    digitalWrite(6, LOW);
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```

Conclusion

This project demonstrates how a PIR motion sensor works with Arduino UNO. Motion detection systems are widely used in security alarms, automatic lighting, smart homes, and industrial automation systems.